

AI & Natural Language Processing Glossary

Domain-Specific Technical English Glossary for SLM Translation Task

#	Term	Definition (English)
1	Tokenization	The process of breaking down a continuous text sequence into smaller meaningful units called tokens (words, subwords, or characters) for model input representation.
2	Small Language Model (SLM)	A lightweight artificial intelligence model, typically containing under 10 billion parameters, optimized for high efficiency and local on-device inference.
3	Large Language Model (LLM)	A deep learning model with tens or hundreds of billions of parameters, trained on massive web-scale text datasets to perform diverse language understanding tasks.
4	Attention Mechanism	A key neural network architecture component that allows models to dynamically assign varying weight and focus to different words in a sequence based on contextual context.
5	Transformer Architecture	A foundational deep learning model layout based entirely on parallel self-attention mechanisms, replacing traditional sequential recurrent architectures.
6	Prompt Engineering	The discipline of designing, refining, and structuring text prompts to guide generative AI models toward accurate and desirable outputs.
7	Knowledge Distillation	A compression technique in machine learning where a compact student model is trained under the supervision of a larger teacher model to mirror its predictive capabilities.
8	Quantization	The technique of converting high-precision neural network numerical weights (e.g. 32-bit floating point) into lower-precision formats (e.g. 8-bit integers) to optimize memory footprint.
9	Vector Embedding	A dense numerical vector representation of words, sentences, or concepts mapped into a continuous high-dimensional space capturing semantic similarity.
10	Context Window	The maximum sequence length of tokens that a language model can retain in active memory during a single generation or inference pass.
11	Hallucination	An unwanted generative AI behavior where the model produces confident, fluent, but factually incorrect or unsupported statements.

1 2	Zero-Shot Learning	The capability of a trained language model to execute complex downstream tasks without explicit prior training examples or task-specific fine-tuning.
1 3	Parameter-Efficient Fine-Tuning (PEFT)	A set of adaptation methods like LoRA that freeze core model weights and tune only a small subset of parameters to efficiently adapt to new domains.
1 4	Temperature	A decoding hyperparameter that scaling probability logits to adjust the randomness, creativity, and diversity of model token output generation.
1 5	Recurrent Neural Network (RNN)	A class of neural networks designed for sequential data processing where connections between nodes form a directed graph along a temporal timeline.